JSS 2 MATHEMATICS LESSON NOTES (THIRD TERM)
WEEK 6:

WEEK 8: STATISTICS I – DATA PRESENTATION

TOPIC: Collecting Data; Organizing Data; Frequency Tables

A. INTRODUCTION TO STATISTICS

Definition:

Statistics is the branch of mathematics concerned with:

- Collection
- Organization
- Presentation
- Analysis
- Interpretation of data

WHY STATISTICS IS IMPORTANT

Statistics helps us to:

1. Make informed decisions
2. Identify trends and patterns

3. Compare data effectively
4. Predict future outcomes
5. Solve real-life problems

REAL-LIFE APPLICATIONS

- **Business:** Sales analysis, profits, customer trends
- **Education:** Exam performance, student tracking
- **Health:** Disease spread, medical research
- **Sports:** Player performance analysis
- **Government:** Census and population planning

B. DATA AND TYPES

Definition:

Data refers to facts, figures, or information collected for analysis.

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1. QUALITATIVE DATA (NON-NUMERICAL)

Describes characteristics or categories.

Examples:

- Eye color (brown, blue, black)
- Gender (male, female)
- Favorite food
- Religion
- Types of transport

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2. QUANTITATIVE DATA (NUMERICAL)

(a) DISCRETE DATA

Countable whole numbers only.

Examples:

- Number of students
- Number of cars
- Number of goals scored
- Number of books

(b) CONTINUOUS DATA

Measured values (can include decimals).

Examples:

- Height (1.65 m)
- Weight (54.3 kg)
- Temperature (32.5°C)
- Time (2.75 hours)

C. DATA COLLECTION METHODS

1. Observation

Watching and recording events.

Example: Counting vehicles on a road.

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2. Questionnaire

Written questions for responses.

Example: Favorite subject survey.

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3. Interview

Face-to-face questioning.

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4. Experiment

Controlled scientific test.

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5. Secondary Data

Existing data from books, internet, reports.

PRIMARY VS SECONDARY DATA

Primary Data:

Collected firsthand.

Examples:

- Students' test scores
- Family size
- Classroom survey

Secondary Data:

Already collected by others.

Examples:

- Census data
- Weather reports
- Research publications

D. RAW DATA

Definition:

Raw data is unorganized data in its original form.

Example:

Test scores:
45, 67, 52, 78, 56, 67, 89, 45, 72, 56

Problem with Raw Data:

- Difficult to interpret
- No clear pattern

- Hard to analyze

E. ORDERED DATA (ARRAY)

Definition:

Arranging data in ascending or descending order.

Example:

Raw:
45, 67, 52, 78, 56

Ascending:
45, 52, 56, 67, 78

Advantages:

- Easy to identify highest/lowest
- Helps find range
- Shows pattern clearly

F. RANGE OF DATA

Range=Highest value−Lowest value

Example:

23, 45, 12, 34, 23

Range:
45 − 12 = 33

G. FREQUENCY DISTRIBUTION

Definition:

Frequency is the number of times a value occurs.

TALLY SYSTEM:

- | = 1 || = 2 ||| = 3 |||| = 4 ~~||||~~ = 5 (cross through four) ~~||||~~| = 6

Example 1: Favorite Fruits

Apple, Orange, Banana, Apple, Mango...

Frequency Table:

Fruit	Tally	Frequency
Apple	𝍸	5
Orange	III	3
Banana	III	3
Mango	II	2

Interpretation:

- Most common: Apple
- Least common: Mango

Example 2: Number of Siblings

Data:
2, 3, 1, 2, 4, 3, 2...

Frequency Table:

Siblings Frequency

Siblings	Frequency
1	3
2	8
3	5
4	4

H. GROUPED FREQUENCY TABLE

When Used:

- Large data set
- Wide range
- Easier interpretation

Example 3: Test Scores

Intervals:
40–49, 50–59, 60–69, 70–79, 80–89

Class Interval	Frequency
40–49	4
50–59	6
60–69	7

Class Interval	Frequency
70–79	5
80–89	3

Observation:

Most students scored 60–69.

I. STEPS TO BUILD FREQUENCY TABLE

1. Collect data
2. Find highest and lowest values
3. Compute range
4. Choose class intervals
5. Make tally marks
6. Count frequencies
7. Check total = number of data items

J. PRACTICAL DATA COLLECTION

From Home:

- Number of rooms
- Daily meals

- TV viewing hours

From School:

- Test scores
- Favorite subjects
- Shoe sizes

From Market:

- Price of goods
- Number of customers
- Product types

CLASSWORK

1. Define statistics
2. What is data?
3. Give examples of qualitative data
4. Differentiate discrete and continuous data
5. What is raw data?
6. Why organize data?
7. Create frequency table from given data

8. What is range?
9. Why use tally marks?
10. What is grouped data?

REVISION SUMMARY

- Statistics = data study
- Data types = qualitative & quantitative
- Quantitative = discrete + continuous
- Raw data = unorganized
- Frequency = occurrence count
- Range = max − min
- Grouping = simplifies large data

ASSIGNMENT: Explain the following :

- qualitative and quantitative data
- discrete and continuous data

- questionnaire

- Arrange in ascending order
- Find range

- Ungrouped tables

- Grouped tables

- Most frequent value
- Least frequent value
- Patterns in data